

Exact Schrödinger Attention on an Oracle Route-Policy Benchmark

A bounded empirical record through D1

Abstract

This paper reports the completed empirical record for an exact, classical implementation of Schrödinger attention (SA): complex attention amplitudes are initialized from softmax probabilities, evolved by a unitary matrix exponential, and measured with squared magnitudes. The scientific question was deliberately modest: under a matched small-model route-policy task, does this primitive learn meaningfully different and better policies than softmax attention? The answer is mixed. Exact amplitude evolution is numerically viable, is not merely inert, and can produce substantially different attention distributions and policies. Both SA and softmax learn the oracle route task. A discovery pair (seed 1701) favored SA at 8,000 equal updates (+2.94 percentage points [pp] valid-route quality; +5.38 pp on an unseen-routine-map cohort), but SA's measured training-core cost was about 3.27 times softmax's. In four fresh paired replications, the frozen primary outcome instead favored softmax by 0.449 pp on average (SA minus softmax 95% descriptive paired-t interval [-3.372, +2.474] pp; one positive pair of four). A small challenge pass@32 difference at the untuned temperature (+2.734 pp) shrank to +0.781 pp after a symmetric, predeclared temperature selection with quality floors; its seed interval was [-5.050, +6.612] pp and only two pairs favored SA. Thus the evidence supports *different learned policies*, not a robust quality, diversity, transfer, or “creativity” advantage. No model-bearing evaluation exists on the reserved route-policy final-test split. This is a reproducible research record and a basis for deciding whether a narrowly powered confirmation is worth authorizing, not evidence for language-model scaling or quantum computation.

1. Question and scope

Softmax attention turns query-key scores into a non-negative distribution. The proposal behind this work was that a latent complex amplitude state could give a network an additional route to reinforcement and cancellation before it is reduced to an ordinary probability distribution. This is a mathematical, classical hypothesis: it does **not** posit quantum hardware, quantum randomness, or a physical quantum state in the model.

The original research proposal set a broad programme spanning synthetic tasks, language modelling, scaling, and causal attention concerns (proposal [repo: ../schrodinger_attention_research_proposal.md]). The completed work is far narrower. It establishes and evaluates a short-sequence, bidirectional route-policy reference model with an exact shortest-path oracle. It does not evaluate a causal language model, long contexts, approximate or fused kernels, energy use, or scaling laws. Those distinctions matter: an exact short-sequence reference can validate a mechanism and expose failure modes, but cannot answer whether the mechanism is economically useful in a transformer deployed at scale.

The central comparison was frozen in internal execution plans: matched softmax versus exact SA on the same 12x12 grid-policy data, architecture, optimizer, updates, within-pair initialization, batch stream, and evaluation draws. The main unit of architecture replication is a paired training seed, not individual states, problems, maps, rollouts, heads, or checkpoints. This paper retains unsuccessful dataset stages and provenance limitations rather than retrospectively presenting only the successful benchmark construction or the favorable discovery seed.

Related work and terminology

The conventional baseline follows the attention framing of Vaswani et al., *Attention Is All You Need*. Recent adjacent preprints include Nahid et al.'s phase-aware score-factorization method, *Q-Interference*, and Reinhardt and Hauser's Born-rule analog for simplex softmax, *A Quantum Roadmap for Softmax Attention*. They are cited as related preprints, not as an exhaustive review or a priority claim. In particular, Born-rule terminology by itself does not establish additional expressivity or a performance benefit. The implementation here is distinct from phase-aware factorization: it explicitly forms a real symmetric Hamiltonian, differentiates through one exact matrix exponential per head/batch score matrix, and applies that evolution to each attention row.

We use “Schrödinger” as a description of the update equation, not as evidence that the computation is quantum. “Novel route” is also deliberately technical: it means a legal shortest route whose canonical action signature is absent from complete training-route and suffix support, after specified spatial and reversal equivalences. It is not a claim about human originality, broad creativity, or unbounded planning.

2. Implemented method

For a head with score matrix $S \in \mathbb{R}^{(L \times L)}$, the actual reference implementation uses

$$H = (S + S^T) / (2\sqrt{L}); U = \exp(-i \Delta t H).$$

It initializes each row-state as

$$\psi_0 = \sqrt{\text{softmax}(S)} \odot \exp(i \gamma S); \psi_1 = \psi_0 U^T; A = |\psi_1|^2.$$

The attended output is AV . The transpose in the row-state update is intentional: each query row is represented as an amplitude state over keys. The Hamiltonian is real symmetric, hence Hermitian after complex conversion, and the matrix exponential is unitary up to numerical precision. Squared complex magnitudes are non-negative and sum to one per row; no post-measurement renormalization is performed. The exact source is [attention.py](#) [repo: [../schrodinger/attention.py](#)] (SHA-256 [e1ce15be10caf4b2165191f9835256602f4f7b8bd11bffa94f676eac40b0fd6a](#)).

Both modes use the same route-policy backbone: 13 tokens (a CLS token plus 12 grid rows), two transformer blocks, two heads, model width 64, feed-forward width 128, and 70,540 parameters. For SA, each head learns bounded Δt and phase scale γ ; for softmax, two learned scales control score and value matching. This equivalence of total parameter count is useful but not a guarantee of equivalent inductive bias or computational cost. The implementation is in [route_policy.py](#) [repo: [../schrodinger/route_policy.py](#)] (SHA-256 [95cb7463693c7ac696d9034eaf16a155c0363f38e819cbe56c00408d49b170c4](#)).

This model is not causal language-model attention. It receives a fully available 13-token grid representation. Legal-action masking is applied only at the final four-action policy distribution: walls and grid edges are masked; the attention mechanism itself is not a causal mask. Accordingly, results cannot be transferred to decoder-only settings, where a symmetric sequence Hamiltonian would create a substantive causal-design problem.

Numerical tests checked Hermiticity, unitarity, Born-row normalization, finite and double-precision gradients, and the $dt=0$ reduction. Across retained runs, maximum unitarity/Born-row error was about $1.2e-6$, below the $2e-3$ guard. The $dt=0$ condition uses the same learned phase and weights but switches evolution time off. It reduces to softmax weighting because the amplitude magnitude is the square root of softmax probabilities; it is not an independently trained softmax model. These implementation and interpretation boundaries are important for every mechanism claim below.

3. Benchmark, oracle, and evaluation protocol

The final accepted benchmark comprises 12x12 obstacle grids with route lengths 14--16 and 16 start/goal problems per map. Training contains 64 homogeneous maps (1,024 problems). Validation contains 24 routine and 8 mixed-composition challenge maps (512 problems total); the reserved, untouched test contains 96 routine and 32 challenge maps (2,048 problems). The exact construction expanded each family candidate pool to 512 maps while retaining the frozen geometry, selection, and novelty rules. Its independent review is recorded in the pool-512 report [repo: [../execution/next_level_v3_pool512/final_report.md](#)] (report SHA-256 1f5e1f99bd847c2f4dc64d31935dcbb88abcc7794f26e468277765e341211d85).

The oracle performs breadth-first search from the goal. At each state the teacher distribution q assigns probability proportional to the number of shortest completions available through each legal optimal action. This avoids pretending that a multi-solution shortest-path problem has one privileged action. The action legality mask excludes wall and off-grid moves only. The metric/oracle code is `route_policy_metrics.py` [repo: [../schrodinger/route_policy_metrics.py](#)] (SHA-256 a471ae2c40973cb15584f924061b3ace24ef5af3b1e31702b79056055f317e34).

For the 8,000-update route comparisons, the saved configuration uses AdamW (learning rate 0.001, betas 0.9/0.999, epsilon $1e-8$, weight decay 0.01), gradient clipping at 1.0, and batches of 64; 8,000 updates therefore equal 512,000 sampled exposures. The route runs use CPU execution with two Torch intra-op and one inter-op thread. The repository pins its relevant packages in `requirements.txt` [repo: [../requirements.txt](#)]: Torch 2.14.0, NumPy 2.5.3, and pytest 9.1.1. This is a protocol description, not an assertion that every historical run used identical physical hardware.

Reported policy metrics include proper-score Brier and $KL(q || p)$, greedy exact shortest-route quality, and sampled route metrics using $K=32$ rollouts. Let Q be the fraction of valid exact-shortest routes among K draws; $\text{pass}@K$ is whether at least one valid route occurs. U_{valid}/K is the number of *distinct* valid routes divided by K . `valid_headroom` is distinct valid routes divided by the exact count M of valid shortest solutions, whereas `coverage` refers to novel-route coverage and uses $M(\text{novel})$. These are different denominators. Entropy is reported descriptively but is not calibration and cannot substitute for a proper score or quality control.

Routine and challenge summaries are map-balanced and then combined at fixed 80/20 weights unless labeled otherwise. Common random uniforms align sampled rollouts within a pair. Equal update count gives equal sampled exposures, not equal wall time or compute: exact matrix exponentiation is intrinsically more expensive in this reference. A seed-level mean is therefore the relevant replication statistic. Confidence intervals below are descriptive small-sample paired-t intervals; four seeds are too few to establish normality, superiority, or equivalence, and 512 problems must not be treated as 512 independent training replications.

4. Chronological empirical record

4.1 Initial classifier screen: suggestive but inadequate

The first XOR/COPY screen retained three paired seeds at 2,000 updates. On its held-out-pair XOR primary endpoint, softmax scored 70.90% and exact SA 72.92%, a mean paired difference of +2.02 pp. The frozen success threshold was +3 pp; neither architecture met the 80% mean validation-XOR adequacy requirement. COPY showed no material regression. Only one seed learned XOR robustly enough to reach the high targets, and disabling evolution changed the primary endpoint by about -0.065 pp on average. These data did not attribute the small difference to evolution or qualify sample efficiency.

The numerical reference passed, but execution provenance invalidated controlled timing and equal-time claims: output-directory reuse and unconfirmed process completion made attempt-specific timing/non-overlap unrecoverable. These results remain descriptive, with their limitation plainly recorded in the initial final report [repo: [../execution/final_report.md](#)]. They were an appropriate reason not to scale a language model, not a negative verdict on the mechanism.

4.2 Same-weights mechanism diagnostic: evolution is heterogeneous

A checkpoint-only normal-versus-dt0 diagnostic then asked a narrower question: with all learned weights fixed, does evolution alter attention and downstream predictions? It does. Across 240 direct attention cells, mean total-variation (TV) redistribution ranged from 0.00553 to 0.15551; only 70 cells satisfied the predeclared small-effect band. In the original primary XOR condition, one seed/layer/head had 15.551% mean CLS-row TV and 18.515% p95. Full-path changes also altered predictions: for one seed, 30 of 512 labels changed with exactly offsetting 15 gains and 15 losses.

Yet there was no consistent useful result. No condition met the frozen noticeable-benefit criterion (mean loss benefit at least 0.01 nats and positive in at least two seeds); gains and harms offset. Direct attention changes, projection changes, and downstream changes differ by head and seed. The correct conclusion is neither “attention is unchanged” nor “interference helped”: it is heterogeneous active evolution without a consistent task benefit in those retained checkpoints (diagnostic report [repo: ../execution/diagnostics/final_report.md]).

4.3 Constructing a benchmark capable of testing route diversity

Several frozen feasibility stages stopped before any learning. An early construction had only 6 of 512 unseen-composition problems with four or more novel shortest routes. Two 8x8 dataset rungs then failed predeclared allocation gates. A first 12x12 pool created useful partial oracle evidence but missed the challenge-test target by seven maps. These are benchmark-construction failures, not model failures; no thresholds were relaxed or favorable map selection made.

The 512-pool 12x12 expansion finally passed all construction gates. Challenge problems each have at least four qualifying structurally novel shortest routes and a specified novel-solution fraction; routine problems have none under the same support definition. Importantly, these are *oracle opportunities*, not model-generated novelty. This succession of stops is evidence that a named “new map” is not enough to support a novelty claim, and it constrains how later route results may be described.

4.4 Initial route training: learning, but softmax was the practical baseline

At 1,000 equal updates on discovery seed 1701, both matched 70,540-parameter models learned relative to initialization. SA's Brier fell from 0.45874 to 0.17502 and KL from 0.83877 to 0.34705; T=1 valid-route Q rose to 20.78%. Softmax was slightly better at that point (Brier 0.172393, Q 21.17%). SA's core training took 15.792 s versus 4.750 s for softmax (3.32x). At matched core time, softmax at update 1,000 was much stronger than SA at update 300: Q 21.17% versus 9.89%.

The planned softmax-only continuation showed a baseline usability limitation: by 8,000 updates, greedy Q was 53.85% and T1 Q 34.23%, below 80%/70% gates. Diagnostic analyses found training-DAG KL improving from 0.204 to 0.101 while aligned routine DAG KL was approximately flat (0.196 to 0.194) and mixed KL worsened (0.568 to 0.667). This is a baseline generalization diagnosis, not an identified architectural cause. A three-cohort view (seen pairs/new goals on seen maps/unseen routine maps) was 54.99/48.33/39.15% T1 success, likewise descriptive and one-seed only. See the training comparison [repo: ../execution/model_training_comparison/final_report.md].

4.5 Discovery pair: a reason to replicate, not a conclusion

Training both models to 8,000 equal updates on seed 1701 changed the picture. SA T1 Q was 37.17% versus softmax 34.23% (+2.94 pp); greedy quality was 56.72% versus 53.85%. On frozen transfer cohorts, SA-minus-softmax Q was +0.48 pp for seen pairs/maps, +2.69 pp for new goals on seen maps, and +5.38 pp for unseen routine maps. Their policies were meaningfully different: action-probability TV was 0.0969, argmax disagreement 11.33%, and many differing actions were both oracle-optimal.

This is intriguing, but not a favorable narrative to generalize. At T=1, SA did not improve distinct valid novel routes per K (0.8789% vs 0.8984%); an SA T=.75 novelty increase was confounded by much higher Q and failed the predeclared quality match. A same-trained SA dt0 probe slightly improved mean CE, so it did not establish evolution as the training-causal source of SA's outcome. Core training cost was 126.85 s for SA versus 38.82 s (3.27x) for softmax. Timing was sequential and descriptive, not a randomized hardware benchmark. The right

action was fresh replication, which was performed (paired-behavior report [repo: [../execution/paired_behavior/final_report.md](#)]).

4.6 Fresh four-pair replication: primary result does not replicate

Fresh seeds 1702--1705 used the same architecture, data, optimizer, 8,000 updates, 512,000 exposures, $T=1/K32$ endpoint, and within-pair initialization and streams. The frozen primary overall valid-route Q was 38.198% for softmax and 37.749% for SA: SA-minus-softmax = -0.449 pp, positive in one of four pairs. The sample SD was 1.837 pp and the descriptive 95% paired interval was [-3.372, +2.474] pp. Routine Q was -0.584 pp and frozen unseen-routine-map C Q was -0.391 pp. Overall greedy Q was -1.549 pp. Thus no replicated overall quality or transfer advantage was demonstrated.

There is a small setting-specific signal worth reporting but not promoting. Challenge pass@32 was 51.367% (softmax) versus 54.102% (SA), +2.734 pp, with three positive pairs. Challenge distinct valid routes/K was 11.176% versus 11.304% (+0.128 pp); exact valid-solution fraction covered was 4.239% versus 4.630% (+0.391 pp). These measures differ: pass@32 is whether a finite bag contains one valid route, whereas valid headroom is route-set coverage. Overall valid-route diversity was slightly lower (-0.189 pp), and strict-novel metrics were near equal. Policy difference persisted (mean state-policy TV 0.1051 and argmax disagreement 11.46%), but different policies are not necessarily better or more diverse within a policy.

Each pair's three SA temperatures (.75/1/1.25) were retained. $T=1$ was nearest the softmax quality point; only seeds 1703 and 1704 met the approximate mixture and stratum quality tolerances, and both had lower overall distinct-valid/K than softmax. No grid was extended or interpolated. Mean core times were 39.867 s softmax and 128.316 s SA, 3.22x. The complete fresh-seed record and definitions are in the replication report [repo: [../execution/seed_replication/final_report.md](#)] (SHA-256 [c5ca3ecf6474eaa09f489225c094400438e58ec9a209a802cdda29ecffc854ed](#)).

4.7 D0/D1 difficult-problem analysis: a reduced, uncertain signal

D0 reused the same four seeds and eight challenge validation maps; it performed no new training or inference. The $T1$ challenge pass@K curve rose from a +0.092 pp difference at $K=1$ to +2.734 pp at $K=32$. Leave-one-map-out differences all remained positive, but map 344 contributed 7 of 14 net problem-seed successes. SA solved more problems at least once (277 vs 263) without a clear increase in route multiplicity on shared solved problems. This is a robustness description of a selected validation panel, not an independent replication or a broad difficult-planning result (D0 report [repo: [../execution/difficult_problem_solving/d0_report.md](#)]).

D1 then performed a frozen symmetric five-temperature search for each mode, retaining all 40 cells. Candidate temperatures had to meet both routine and challenge single-draw Q floors: at least paired softmax $T=1$ minus 2 pp. These are eligibility floors, not exact quality matching. SA seed 1702 at $T=1$ missed the routine floor and SA seed 1705 at $T=1$ missed the challenge floor; their eligible SA selections were $T=.75$. The selected result was challenge pass@32 51.367% softmax versus 52.148% SA (+0.781 pp), positive two of four pairs. Its paired seed interval was [-5.050, +6.612] pp; the conditional reused-panel map-bootstrap interval was [-1.758, +3.711] pp. Both cross zero. Challenge distinct-valid/K was 11.176% versus 11.249% (+0.073 pp), and valid-solution coverage 4.239% versus 4.623%.

This D1 result is quality-controlled only in the limited floor sense, reuses the same validation maps for selection and scoring, and does not equalize training compute. It weakens rather than confirms the raw $T1$ signal. SA's route variation was mixed: slightly higher coverage and U_{valid} , but higher duplicate rate and nonuniform per-seed directions. The audited report correctly records D2 forecast as deferred/no-go, not as a model-runtime failure. D1 report [repo: [../execution/difficult_problem_solving/d1-report.md](#)], SHA-256 [7c56622bd375e0dc8f66cce09ae126e428728ad703b04010ce94737af3874d84](#); the independent audit is here [repo: [../execution/difficult_problem_solving/d1-result-audit.md](#)], SHA-256 [ada20693ec3edb8381fc02948b645227263aa539f25a8a650fbbc81c6997cfec](#).

4.8 Consolidated numerical results

These rows are not a pooled meta-analysis. The discovery pair selected the subsequent investigation; the fresh four-pair result is the frozen primary replication estimate; D0 reuses the fresh validation result; and D1 reuses the same validation panel after frozen floor-based temperature selection. Q is single-draw valid-shortest-route quality; pass@32 is the fraction of problems with at least one valid shortest route among 32 draws.

Status and endpoint	Discovery 1701, 8k overall T1 Q
Softmax	34.23%
Exact SA	37.17%
SA – softmax	+2.94 pp
Unit / interpretation	One seed; exploratory discovery

Status and endpoint	Fresh 1702–1705, 8k overall T1 Q (primary)
Softmax	38.198%
Exact SA	37.749%
SA – softmax	–0.449 pp
Unit / interpretation	Four paired seeds; descriptive 95% interval [-3.372, +2.474] pp

Status and endpoint	Fresh 1702–1705, challenge T1 pass@32
Softmax	51.367%
Exact SA	54.102%
SA – softmax	+2.734 pp
Unit / interpretation	Four paired seeds; secondary; reused by D0

Status and endpoint	D1 floor-selected challenge pass@32
Softmax	51.367%
Exact SA	52.148%
SA – softmax	+0.781 pp
Unit / interpretation	Same validation panel; exploratory; four-seed interval [-5.050, +6.612] pp

5. Discussion

The most durable positive finding is methodological: exact amplitude evolution can be trained stably in a small conventional model and can change attention, actions, route solutions, and learning trajectories. This defeats the narrow explanation that the implementation simply reproduces softmax or that evolution is numerically negligible. It does not identify which aspects of phase, Hamiltonian coupling, additional learned controls, optimization, or finite-task geometry create those differences.

The quality result is not favorable. The discovery pair justified replication; the fresh primary result did not reproduce it. The challenge coverage signal is smaller after symmetric temperature selection, has wide seed uncertainty, and comes from a repeatedly analyzed eight-map validation panel. It is compatible with a small useful effect, no effect, or a harmful effect of practical size. The same data cannot establish equivalence because the interval is wide. A claim that SA increases “creativity,” useful uncertainty, general transfer, or difficult-problem solving would exceed the evidence.

Cost changes the decision threshold. Even if a small advantage existed, a roughly 3.22x core-training cost in this exact CPU reference means it would need substantially stronger quality, sample-efficiency, or deployment value to be competitive. These measurements are not estimates for GPU kernels, approximate evolution, or long sequences, but they rule out a claim that the implemented exact reference is already cost-neutral. Historical classifier timing is invalid and must not be combined with later timing. The current qualified operational debit is 4764.416447001050 of 7200 seconds; it includes a 300-second administrative uncertainty allowance and is not a measured physical-total runtime. Cumulative budgets from different reports must not be summed.

Important limitations remain. The benchmark is small and selected; the initial route data has an all-invalid within-map ordering limitation, though source/order bindings and legal-route checks support retained alignment. The teacher is an oracle for shortest paths, not a natural-language target. Temperature selection and difficult-problem scoring reuse validation data. The baseline itself has meaningful held-out generalization limits. Mechanism tests are same-weights interventions, not independently trained causal ablations. Finally, no model-bearing evaluation has used the reserved route-policy final-test split. These constraints are reasons to preserve the result as inconclusive, not excuses to select a positive metric.

6. Conclusion

Exact Schrödinger attention is a numerically sound, deterministic classical attention primitive that changes learned behavior. In this bounded benchmark, it did not demonstrate a robust replicated benefit over matched softmax. The favorable 1701 outcome failed to replicate on the frozen primary endpoint, and the residual difficult-panel coverage difference is small and uncertain after limited quality control. The implemented reference is materially more expensive in core training. The justified conclusion is therefore: **do not scale or make broad capability claims on the present evidence**. A future study should either stop here or first meet a strict authorization, runtime-forecast, untouched-map, and replication standard described in next research steps [repo: next_research_steps.md].

Appendix A. Reproducibility map

This paper describes repository artifacts; an external reader needs a snapshot of this repository to resolve the relative links and reproduce local hashes.

Evidence	Original hypothesis and planned controls
Immutable/reviewable location	proposal [repo: ../schrodinger_attention_research_proposal.md]
Hash or verification boundary	Historical proposal, not a claim that all stages ran
Evidence	Exact attention source
Immutable/reviewable location	schrodinger/attention.py [repo: ../schrodinger/attention.py]
Hash or verification boundary	e1ce15be10caf4b2165191f9835256602f4f7b8bd11bffa94f676eac40b0fd6a
Evidence	Matched policy source
Immutable/reviewable location	schrodinger/route_policy.py [repo: ../schrodinger/route_policy.py]
Hash or verification boundary	95cb7463693c7ac696d9034eaf16a155c0363f38e819cbe56c00408d49b170c4

Evidence	Oracle and metrics
Immutable/reviewable location	<code>schrodinger/route_policy_metrics.py</code> [repo: <code>../schrodinger/route_policy_metrics.py</code>]
Hash or verification boundary	<code>a471ae2c40973cb15584f924061b3ace24ef5af3b1e31702b79056055f317e34</code>
Evidence	Initial screen and timing limitation
Immutable/reviewable location	execution final report [repo: <code>../execution/final_report.md</code>]
Hash or verification boundary	Retained descriptive endpoints; equal-time invalid
Evidence	Same-weights diagnostic
Immutable/reviewable location	diagnostic final report [repo: <code>../execution/diagnostics/final_report.md</code>]
Hash or verification boundary	19 diagnostic / 67 total tests reported
Evidence	Complete 12x12 dataset
Immutable/reviewable location	pool-512 report [repo: <code>../execution/next_level_v3_pool512/final_report.md</code>]
Hash or verification boundary	<code>1f5e1f99bd847c2f4dc64d31935dcbb88abcc7794f26e468277765e341211d85</code>
Evidence	First route pair and baseline stop
Immutable/reviewable location	model comparison [repo: <code>../execution/model_training_comparison/final_report.md</code>]
Hash or verification boundary	Independent final audit linked in report
Evidence	Discovery-pair behavior
Immutable/reviewable location	paired behavior [repo: <code>../execution/paired_behavior/final_report.md</code>]
Hash or verification boundary	Independent auxiliary audit linked in report
Evidence	Fresh four-seed primary result
Immutable/reviewable location	seed replication [repo: <code>../execution/seed_replication/final_report.md</code>]
Hash or verification boundary	<code>c5ca3ecf6474eaa09f489225c094400438e58ec9a209a802cdda29ecffc854ed</code>
Evidence	D0 reused-panel screen
Immutable/reviewable location	D0 report [repo: <code>../execution/difficult_problem_solving/d0_report.md</code>]
Hash or verification boundary	No new training/inference

Evidence	D1 selected-temperature result
Immutable/reviewable location	D1 report [repo: ../execution/difficult_problem_solving/d1-report.md]
Hash or verification boundary	7c56622bd375e0dc8f66cce09ae126e428728ad703b04010ce94737af3874d84

Evidence	D1 independent result/accounting review
Immutable/reviewable location	D1 audit [repo: ../execution/difficult_problem_solving/d1-result-audit.md]
Hash or verification boundary	ada20693ec3edb8381fc02948b645227263aa539f25a8a650fbbc81c6997cfec

The output manifests, attempts, ledgers, plans, and independent reviews linked from these reports provide artifact-level hashes and command/provenance records. They do not turn a small, reused validation analysis into a final-test result.

Appendix B. Interpretation rules used in this paper

This appendix makes explicit several distinctions that are easy to lose when a compact result table is read in isolation.

Equal updates, equal data exposure, and equal compute are different. Within each paired run the two models share initialization, minibatch digests, update count, optimizer settings, and decoding uniforms. That is a strong control for one kind of comparison: whether the architectures take different learning paths under equal update exposure. It is not a control for computation. Exact SA constructs a complex matrix exponential per head/batch score matrix and was slower in the measured CPU reference. The reported 3.22x mean core-time ratio from fresh replications is therefore a relevant practical finding, even though it is not a universal estimate for optimized GPU kernels. Conversely, a comparison that gives SA fewer updates to match core seconds is a compute-efficiency comparison, not an equal-exposure comparison. Neither should be relabeled as the other.

Route quality and route variety are different. Q counts valid shortest routes in finite samples. Pass@32 asks whether a problem receives at least one such route. A model can increase pass@32 by spreading rare success across more problems without generating many different solutions on problems it already solves. That is the likely interpretation of the D0 pattern: SA had more problem-seed successes but no convincing increase in distinct route count among the shared solved subset. U_{valid}/K corrects for repeated valid outputs but still has K in its denominator; valid headroom instead normalizes by the exact number of shortest routes available for that problem. Neither establishes an ability to discover new strategies. Strict novelty further requires absence from the complete canonical training support, and it was supplementary in the route study.

Temperature floors are not matching. D1 selected the temperature that maximized challenge pass@32 among candidates meeting routine and challenge Q floors. Its design protects against accepting an endpoint that visibly loses single-draw route quality, but a candidate can be materially better or worse in either stratum while remaining eligible. Its result should therefore be called “floor-controlled” or “limited quality-controlled,” never exactly quality-matched. Moreover the same validation panel chose and evaluated the temperatures, so the D1 point estimate is exploratory even though every cell and selection rule was frozen before the new endpoints were computed.

A $\Delta t=0$ intervention is not an independently trained baseline. Given the implementation, $\Delta t=0$ yields $|\sqrt{\text{softmax}(S)} \exp(i \gamma S)|^2 = \text{softmax}(S)$ exactly. The learned phase is invisible at measurement without evolution. Thus a same-checkpoint $\Delta t=0$ run asks what happens when evolution is removed from an already trained SA model; it cannot determine whether independently training without evolution would reach the same weights, optimization basin, or route policy. It also means a proposed “phase-only, no-evolution” model would be degenerate with softmax in this exact formulation. Meaningful causal ablations must be independently trained and account for capacity and compute.

The statistical unit is a seed. A single seed emits many route rows, but those rows share weights and correlated map geometry. The four fresh pairs give four architecture replications. The paired intervals are intentionally shown to communicate imprecision, not to create a pass/fail ritual. The D1 map bootstrap answers a different conditional question—variation over the eight already selected maps at fixed seed/temperature selection—and does not repair the lack of fresh training seeds or panel reuse. Similarly, leave-one-map-out positivity tells us an average is not caused by one indispensable map, not that the result generalizes to new maps.

“Classical” is a substantive boundary. All operations here are conventional PyTorch-like complex tensor arithmetic. Unit norm is a numerical invariant and the Born rule is a deterministic map from complex amplitudes to real weights. Sampling occurs later from an ordinary policy distribution. No claim in this paper rests on quantum hardware, measurement randomness, or a physical interpretation of the learned amplitudes.

Appendix C. Stage-by-stage decision ledger

Stage	Initial XOR/COPY screen
Decision made by the evidence	Exact module was viable; small retained difference did not meet gate; timing comparison invalid
What it did not decide	General SA efficacy or efficiency
Stage	Same-weights diagnostic
Decision made by the evidence	Evolution can substantially redistribute attention and alter predictions
What it did not decide	That evolution causes a training benefit
Stage	Early feasibility ladders
Decision made by the evidence	Several frozen constructions lacked adequate novel-route support or quotas
What it did not decide	That models fail at novelty/diversity
Stage	512-pool benchmark
Decision made by the evidence	A complete oracle-controlled route dataset exists
What it did not decide	That either architecture learns it well
Stage	1k paired training
Decision made by the evidence	Both learn; softmax is more cost-effective at measured equal time
What it did not decide	Long-run SA quality or transfer
Stage	1701 discovery pair
Decision made by the evidence	A replicate-worthy SA advantage is possible in one seed
What it did not decide	Replicated advantage or creativity
Stage	Fresh 1702--1705 pairs
Decision made by the evidence	Primary overall advantage did not replicate
What it did not decide	Equivalence or impossibility of a small effect

Stage	D0
Decision made by the evidence	Raw challenge coverage is not solely a single-map artifact
What it did not decide	Independent difficult-task evidence

Stage	D1
Decision made by the evidence	Temperature floors reduce the raw signal to a small, uncertain contrast
What it did not decide	Exact quality matching, final confirmation, or D2 authorization

The operational resource ledger has a different role from this scientific ledger. It preserves attempted work, administrative allowances, and provenance without rewriting history. Qualified totals are useful for authorization and audit; they are not a substitute for the measured core timing fields used for a compute comparison. In particular, the final D1 global debit includes an explicit historical uncertainty allowance. It would be misleading to describe it as all physical experiment runtime or to add it to intermediate cumulative totals that already include inherited debits.

Appendix D. Minimum reporting requirements for any successor study

A successor study can be smaller than a language-model project while still being more decisive than the current record. At minimum it should publish: the exact source revision and hashes; a frozen model/configuration table; train/tune/ confirmation-map identities and disjointness checks; seed and model-order rules; all failed and completed attempt owners; measured core, inference, and overhead timings separately; one primary endpoint and its practical threshold; temperature selection separated from confirmation scoring; seed-level raw paired outcomes; map-aware sensitivity summaries; all eligible temperatures rather than only the winner; and a clear statement of whether final test access occurred.

The comparison should also reserve the right to be negative. A sensible precommitment is that a failed confirmation will be written up as a bounded negative result, rather than triggering a hidden sequence of map changes, temperature grids, checkpoint choices, or longer training. This is especially important because the current study already contains a plausible discovery seed and a plausible challenge subgroup. Those are precisely the situations in which selection can manufacture a narrative without producing a dependable effect.

Appendix E. Why the exact reference is useful despite its cost

The exact matrix exponential is not proposed as a production attention kernel. Its value in this programme is epistemic. It gives a well-defined target calculation: for a symmetric real score-derived Hamiltonian, the unitary is computed directly rather than replaced by a truncation whose stability or error might itself explain a performance difference. It also makes several numerical claims testable. Hermiticity, unitarity, normalization, and the $dt=0$ reduction have crisp expected values. A later efficient approximation would need to show not only an end-task result, but also how closely it preserves the reference operation and whether any change in quality is due to approximation rather than the amplitude mechanism.

There is a complementary limitation. The reference's costs grow unfavorably as the attention-state length increases, and complex matrix operations have different implementation characteristics from softmax. A result on 13 tokens cannot establish latency, memory, prefill throughput, decode throughput, KV-cache behavior, accelerator utilization, or energy for language models. The 3.22x training-core figure should therefore be read narrowly: it is an observed economic disadvantage of the exact CPU implementation on this task. It neither licenses a claim that all unitary/phase methods are 3.22x slower nor permits an efficient variant to inherit a benefit that the exact reference has not shown.

The same reasoning applies to causality. For every query in this benchmark, the full grid is a legitimate input. A decoder sequence instead exposes only a prefix. Simply masking a score matrix while symmetrizing it can break

either normal causal semantics or the Hermitian structure on which unitary evolution relies. A causal formulation may need a query-specific key-subspace evolution or a different factorization. That is an architectural research problem, not a minor engineering port. It should be solved and numerically audited before any claim that the present exact method improves autoregressive attention.

Appendix F. Reading the discovery and replication together

The 1701 result should not be discarded: it revealed that an equal-update SA learning path can be better than softmax on some instances and supplied a concrete candidate outcome for replication. But it cannot be pooled with seeds 1702--1705 as though all five were a prospectively homogeneous sample. The discovery seed informed the decision to examine transfer cohorts, temperatures, and difficult-panel coverage; including it in the primary fresh-replication mean would understate the selection process. Reporting it separately is the more informative choice: it tells future investigators what signal motivated the work while preserving the conclusion that the signal was not reliably reproduced.

The fresh seed trajectories add nuance without changing that conclusion. SA held a mean early Q advantage at 1,000 updates (28.096% versus 27.383%) that faded by 4,000 and reversed at 8,000. This pattern is hypothesis-generating, not evidence that a favorable early checkpoint should replace the frozen 8,000 endpoint. It could reflect different optimization dynamics, overfitting, sampling noise, or details of the route representation. It becomes scientifically useful only if a future study predeclares an early-stage endpoint and demonstrates it under equal compute and fresh seeds.

Finally, the small D1 difference is not inconsistent with a negative overall-Q mean: the endpoints ask different questions and weight different subsets. A model can have a modestly higher chance of one success in a difficult finite sampling bag while being no better, or worse, in single-draw quality and overall route validity. That possibility is why the paper reports both measures and refuses to choose the favorable one as a universal score. The open question is not whether a positive number can be found in the record; it is whether a precommitted, practical advantage remains when new seeds, new maps, matched quality, and cost are all brought to bear.